

Cloud LearnX

Mathematics

Mixed Algebra and Geometry — Practice Worksheet

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Date: 30 July 2026

Topic: Mixed Algebra and Geometry

Instructions:

- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

Answer ALL ALL questions.
Write your answers in the spaces provided.
You must write down all the stages in your working.

- 1** A gaming statistics app tracks the number of kills (x) and deaths (y) recorded by a Valorant player across several matches.

The total combined score for two different scoring methods is given by the expression:

$$5x + 3y - 2x + 7y$$

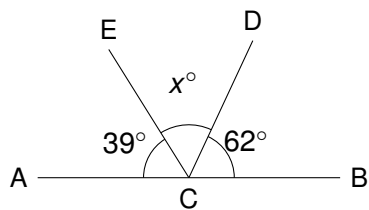
Simplify this expression fully.

.....

(2)

(Total for Question 1 is 2 marks)

2 The diagram shows three angles on a straight line AB , meeting at point C .



AB is a straight line.

Work out the size of angle x .

$x = \dots\dots\dots^\circ$
(2)

(Total for Question 2 is 2 marks)

3 Expand and simplify.

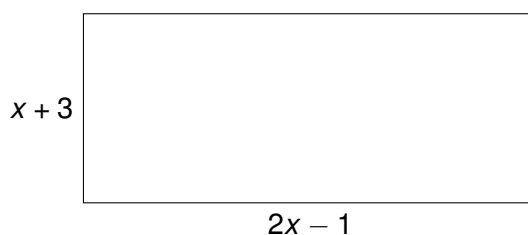
$$3(2x + 5) - 2(x - 4)$$

.....

(2)

(Total for Question 3 is 2 marks)

- 4 A school science lab has a rectangular bench, as shown in the diagram below. The length of the bench is $(2x - 1)$ metres and the width is $(x + 3)$ metres.



- (a) Show that the area, $A \text{ m}^2$, of the bench is given by

$$A = 2x^2 + 5x - 3$$

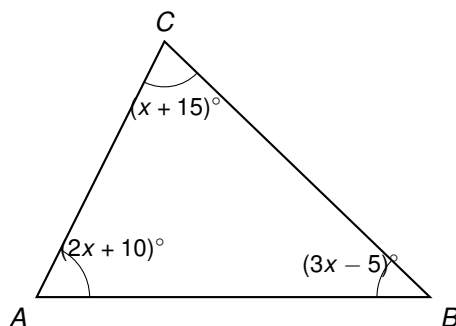
(2)

- (b) Given that $x = 6$, work out the area of the bench.

..... m^2
(1)

(Total for Question 4 is 3 marks)

5 The diagram shows a triangle with angles $(2x + 10)^\circ$, $(3x - 5)^\circ$ and $(x + 15)^\circ$.



(a) Find the value of x .

$x = \dots\dots\dots$
(2)

(b) Work out the size of the smallest angle in the triangle.

$= \dots\dots\dots^\circ$
(1)

(Total for Question 5 is 3 marks)

6 In a mobile game tournament, the total number of points P scored by a team is modelled by the expression

$$P = x^2 + 7x + 12$$

where x is the number of matches won in the bonus round.

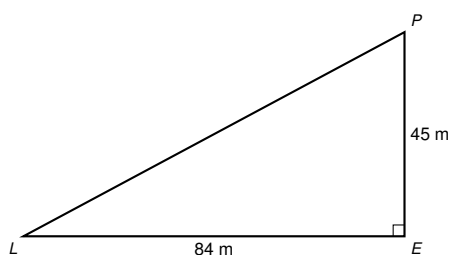
Factorise $x^2 + 7x + 12$ completely.

.....

(3)

(Total for Question 6 is 3 marks)

- 7 A science drone takes off from a launch point L and flies due east to a point E , a distance of 84 m. It then turns and flies due north to a data-collection point P , a distance of 45 m, as shown in the diagram.



The drone must then fly directly back from P to L in a straight line to return to base before its battery runs out.

Work out the direct distance PL . Give your answer correct to 1 decimal place.

..... m

(3)

(Total for Question 7 is 3 marks)

8 A mobile game designer is modelling the in-game currency earned by players. The number of coins earned, related to a difficulty setting x , satisfies the equation:

$$2x^2 + 5x - 3 = 0$$

Use the quadratic formula to solve this equation.

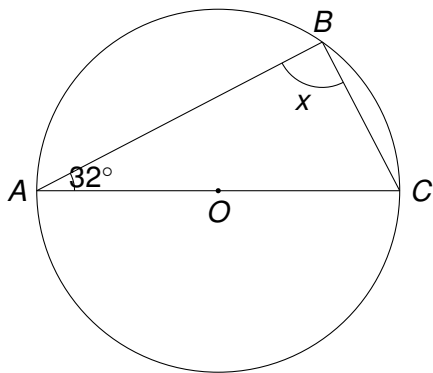
Give your solutions correct to 2 decimal places.

.....

(4)

(Total for Question 8 is 4 marks)

9 A , B and C are points on the circle, with centre O . AC is a diameter of the circle.



Angle $CAB = 32^\circ$.

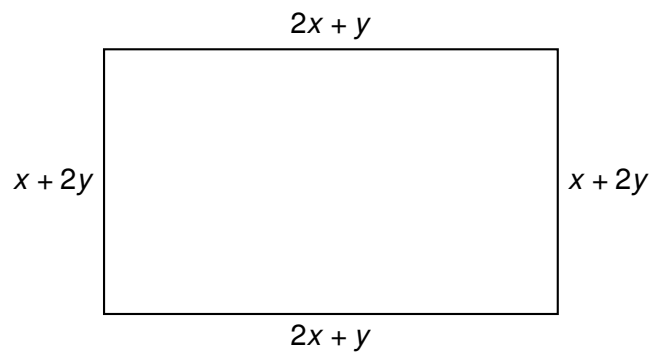
Work out the size of angle x , marked on the diagram. Give a reason for your answer.

$x = \dots\dots\dots^\circ$

(2)

(Total for Question 9 is 2 marks)

- 10** A rectangular greenhouse frame is being built in a garden. The frame is in the shape of a rectangle with length $(2x + y)$ metres and width $(x + 2y)$ metres, as shown in the diagram below.



The perimeter of the greenhouse frame is 42 metres.

The length of the greenhouse is 3 metres longer than the width.

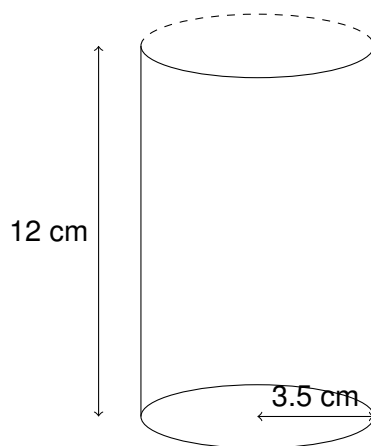
Form two equations in x and y , and solve them simultaneously to find the value of x and the value of y .

.....

(4)

(Total for Question 10 is 4 marks)

- 11** A science laboratory uses a cylindrical container to store a chemical sample. The container is in the shape of a cylinder with radius 3.5 cm and height 12 cm, as shown in the diagram.

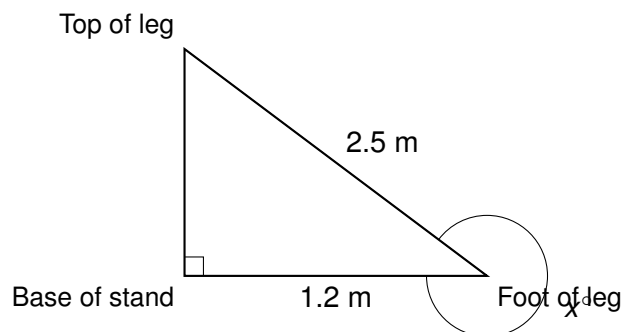


Calculate the volume of the container.
Give your answer to 3 significant figures.

..... cm³
(3)

(Total for Question 11 is 3 marks)

- 12** A streamer sets up a telescope on a tripod to livestream an e-sports viewing party of a meteor shower. The tripod leg forms a right-angled triangle with the ground, as shown in the diagram. The foot of the tripod leg is 1.2 m from the base of the vertical telescope stand, and the tripod leg itself is 2.5 m long.



Work out the size of the angle of elevation, x , that the tripod leg makes with the ground.
Give your answer correct to 1 decimal place.

$$x = \dots\dots\dots^\circ$$

(3)

(Total for Question 12 is 3 marks)

- 13** A mobile game designer is calculating scoring bonuses. A player's bonus score depends on a level number x , and is calculated using two expressions:

$$(2x + 3)^2 \quad \text{and} \quad (2x - 3)^2$$

Show that

$$(2x + 3)^2 - (2x - 3)^2 = 24x$$

(3)

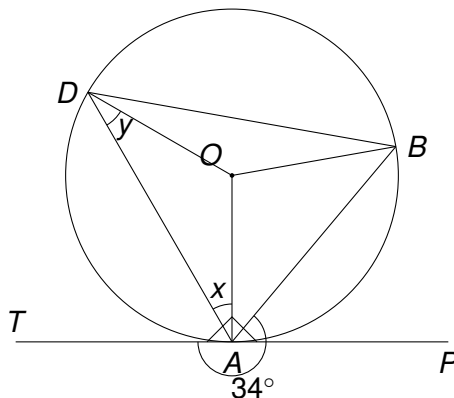
(Total for Question 13 is 3 marks)

14 A , B and D are points on a circle, centre O . TAP is a tangent to the circle at A . BD is a chord, and O is joined to B and to D .

Angle $OAT = 90^\circ$ (angle between tangent and radius).

Angle $BAT = 34^\circ$.

$OB = OD$ (radii of the circle).



(a) Given that angle $OAB = 56^\circ$, work out the size of angle OAD , marked x on the diagram.

$x = \dots\dots\dots^\circ$
(2)

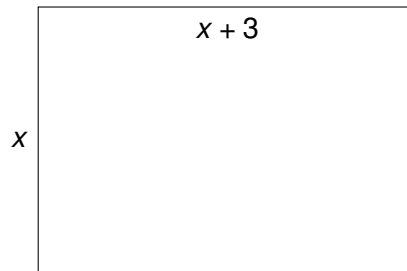
(b) Triangle OAD is isosceles, since $OA = OD$ (radii).

Using your answer to part (a), work out the size of angle ADO , marked y on the diagram.

$y = \dots\dots\dots^\circ$
(2)

(Total for Question 14 is 4 marks)

15 A rectangular solar panel has width x cm and length $(x + 3)$ cm, as shown in the diagram.



The area of the solar panel is 180 cm^2 .

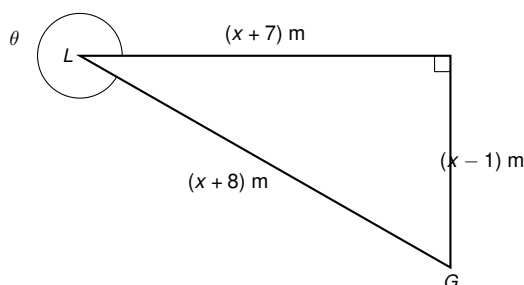
(a) Show that $x^2 + 3x - 180 = 0$. **(2)**

(b) Solve the equation $x^2 + 3x - 180 = 0$ to find the width of the solar panel, giving reasons for rejecting any solution that is not valid. cm
(2)

(Total for Question 15 is 4 marks)

- 16** In a Valorant tournament, a drone camera follows the flight path of a thrown grenade, which is modelled as a straight line from launch point L to landing point G .

The horizontal distance travelled is $(x + 7)$ metres, the vertical height dropped is $(x - 1)$ metres, and the straight-line flight path LG has length $(x + 8)$ metres, as shown in the diagram.



- (a) Show that $x^2 - 30x - 16 = 0$. (2)

- (b) Find the value of x , giving your answer to 3 significant figures.

$x = \dots\dots\dots$
(2)

- (c) Calculate the size of angle θ , the angle of descent of the grenade below the horizontal, giving your answer to 1 decimal place.

$\theta = \dots\dots\dots^\circ$
(1)

(Total for Question 16 is 5 marks)

TOTAL FOR PAPER IS 50 MARKS